

Page 1

$\mathcal{X}$ (all states)

$\mathcal{X}_s$ (all safe states)

$\mathcal{X}_u$ (all unsafe states)

Let $x \in \mathcal{X}_s$.

$$B(x) \geq 0 \quad \forall x \in \mathcal{X}_s \quad (1)$$

$$B(x) < 0 \quad \forall x \in \mathcal{X}_u \quad (2)$$

$$\dot{B}(x) \geq -\alpha(B(x)) \quad \forall x \in \mathcal{X} \quad (3)$$

Boundary conditions:

$$x \in \mathcal{X}_s \Rightarrow B(x) = 0 \quad (4)$$

$$x \in \mathcal{X}_u \Rightarrow B(x) = 1 \quad (5)$$

Loss function:

$$\mathcal{L} = \sum_{x \in \mathcal{X}_s} \max(-B(x), 0) \quad (6)$$

$$+ \sum_{x \in \mathcal{X}_u} \max(B(x), 0) \quad (7)$$

$$+ \sum_{x \in \mathcal{X}} \max\left(-\left(\dot{B}(x) + \alpha(B(x))\right), 0\right) \quad (8)$$

Binary Collision Classifier:

$$\sum_{x \in \mathcal{X}_s} (B(x) - 0) + \sum_{x \in \mathcal{X}_u} (B(x) - 1) \quad (9)$$

Binary cross-entropy style term:

$$y \log y + (1 - y) \log(1 - y) \quad (10)$$

Page 2

Time derivative of B :

$$\frac{dB}{dt} = \frac{\partial B}{\partial x} \cdot \frac{dx}{dt} \quad (11)$$

$$= \nabla_x B \cdot \dot{x} \quad (12)$$

System dynamics:

$$\dot{x} = f(x, a) \quad (13)$$

Finite difference approximation:

$$\frac{\partial B}{\partial t} \approx \frac{B(x_{t+1}) - B(x_t)}{\Delta t} \quad (14)$$

Constraint:

$$\dot{B} \geq -\alpha(B) \quad (15)$$

Discretized form:

$$B(z_{t+1}) - B(z_t) \geq -\alpha B(z_t) \Delta \quad (16)$$

Rearranging:

$$B(z_{t+1}) = B(z_t) - \alpha \Delta B(z_t) \quad (17)$$

$$= (1 - \alpha \Delta) B(z_t) \quad (18)$$

Ideal next state:

$$z_{t+1} = B^{-1}((1 - \alpha \Delta) B(z_t)) \quad (19)$$

Actual system evolution:

$$z_{t+1} = f(z_t) \quad (20)$$